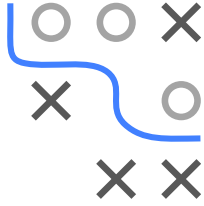
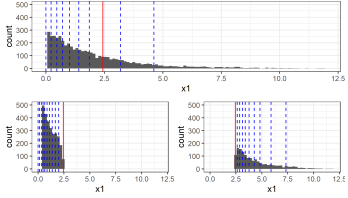
Supervised Learning



Boosting
Gradient Boosting: Deep Dive XGBoost Optimization

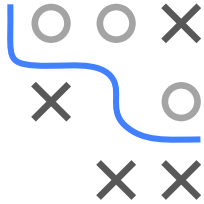


Learning goals

- Understand details of the regularized risk in XGBoost
- Understand approximation of loss used in optimization
- Understand split finding algorithm

RISK MINIMIZATION I

XGBoost uses a risk function with 3 regularization terms:



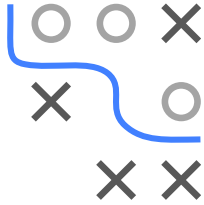
$$\mathcal{R}_{\text{reg}}^{[m]} = \sum_{i=1}^{n} L\left(y^{(i)}, f^{[m-1]}(\boldsymbol{x}^{(i)}) + b^{[m]}(\boldsymbol{x}^{(i)})\right) + \lambda_1 J_1(b^{[m]}) + \lambda_2 J_2(b^{[m]}) + \lambda_3 J_3(b^{[m]}),$$

with $J_1(b^{[m]}) = T^{[m]}$ the number of leaves in the tree to penalize tree depth.

$J_2(b^{[m]}) = \left\|\mathbf{c}^{[m]}\right\|_2^2$ and $J_3(b^{[m]}) = \left\|\mathbf{c}^{[m]}\right\|_1$ are $L2$ and $L1$ penalties of the terminal region values $c_t^{[m]}$, $t = 1, \ldots, T^{[m]}$.

We define $J(b^{[m]}) := \lambda_1 J_1(b^{[m]}) + \lambda_2 J_2(b^{[m]}) + \lambda_3 J_3(b^{[m]})$.

RISK MINIMIZATION II



To approximate the loss in iteration m, a second-order Taylor expansion around $f^{[m-1]}(\boldsymbol{x})$ is computed:

$$L(y, f^{[m-1]}(\boldsymbol{x}) + b^{[m]}(\boldsymbol{x})) \approx$$
$$L(y, f^{[m-1]}(\boldsymbol{x})) + g^{[m]}(\boldsymbol{x})b^{[m]}(\boldsymbol{x}) + \frac{1}{2}h^{[m]}(\boldsymbol{x})b^{[m]}(\boldsymbol{x})^2,$$

with gradient

$$g^{[m]}(\boldsymbol{x}) = \frac{\partial L(y, f^{[m-1]}(\boldsymbol{x}))}{\partial f^{[m-1]}(\boldsymbol{x})}$$

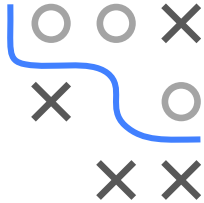
and Hessian

$$h^{[m]}(\boldsymbol{x}) = \frac{\partial^2 L(y, f^{[m-1]}(\boldsymbol{x}))}{\partial f^{[m-1]}(\boldsymbol{x})^2}.$$

Note: $g^{[m]}(\boldsymbol{x})$ are the negative pseudo-residuals $-\tilde{r}^{[m]}$ we use in standard gradient boosting to determine the direction of the update.

RISK MINIMIZATION III

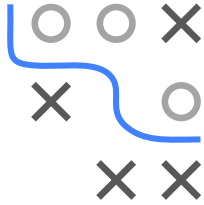
Since $L(y, f^{[m-1]}(\boldsymbol{x}))$ is constant, the optimization simplifies to



$$\begin{aligned}
\mathcal{R}_{\text{reg}}^{[m]} &= \sum_{i=1}^{n} g^{[m]}(\boldsymbol{x}^{(i)}) b^{[m]}(\boldsymbol{x}^{(i)}) + \frac{1}{2} h^{[m]}(\boldsymbol{x}^{(i)}) b^{[m]}(\boldsymbol{x}^{(i)})^2 + J(b^{[m]}) + const \\
&\propto \sum_{t=1}^{T^{[m]}} \sum_{\boldsymbol{x}^{(i)} \in R_t^{[m]}} g^{[m]}(\boldsymbol{x}^{(i)}) c_t^{[m]} + \frac{1}{2} h^{[m]}(\boldsymbol{x}^{(i)}) (c_t^{[m]})^2 + J(b^{[m]}) \\
&= \sum_{t=1}^{T^{[m]}} G_t^{[m]} c_t^{[m]} + \frac{1}{2} H_t^{[m]} (c_t^{[m]})^2 + J(b^{[m]}).
\end{aligned}$$

Where $G_t^{[m]}$ and $H_t^{[m]}$ are the accumulated gradient and Hessian values in terminal node t.

RISK MINIMIZATION IV



Expanding $J(b^{[m]})$:

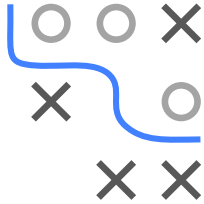
$$\begin{aligned}\mathcal{R}_{\text{reg}}^{[m]} &= \sum_{t=1}^{T^{[m]}} \left(G_t^{[m]} c_t^{[m]} + \frac{1}{2} H_t^{[m]} (c_t^{[m]})^2 + \frac{1}{2} \lambda_2 (c_t^{[m]})^2 + \lambda_3 |c_t^{[m]}| \right) + \lambda_1 T^{[m]} \\ &= \sum_{t=1}^{T^{[m]}} \left(G_t^{[m]} c_t^{[m]} + \frac{1}{2} (H_t^{[m]} + \lambda_2) (c_t^{[m]})^2 + \lambda_3 |c_t^{[m]}| \right) + \lambda_1 T^{[m]}.\end{aligned}$$

Note: The factor $\frac{1}{2}$ is added to the *L2* regularization to simplify the notation as shown in the second step. This does not impact estimation since we can just define $\lambda_2 = 2\tilde{\lambda}_2$.

RISK MINIMIZATION V

Computing the derivative for a terminal node constant value $c_t^{[m]}$ yields

$$\frac{\partial \mathcal{R}_{\text{reg}}^{[m]}}{\partial c_t^{[m]}} = (G_t^{[m]} + \text{sign}\,(c_t^m)\lambda_3) + (H_t^{[m]} + \lambda_2)c_t^m.$$



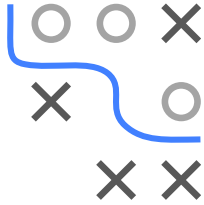
The optimal constants $\hat{c}_1^{[m]}, \ldots, \hat{c}_{T^{[m]}}^{[m]}$ can then be calculated as

$$\hat{c}_t^{[m]} = -\frac{t_{\lambda_3}\left(G_t^{[m]}\right)}{H_t^{[m]} + \lambda_2}, t = 1, \ldots T^{[m]},$$

with

$$t_{\lambda_3}(x) = \begin{cases} x + \lambda_3 & \text{for } x < -\lambda_3 \\ 0 & \text{for } |x| \leq \lambda_3 \\ x - \lambda_3 & \text{for } x > \lambda_3. \end{cases}$$

LOSS MINIMIZATION - SPLIT FINDING I



To evaluate the performance of a candidate split that divides the instances in region $R_t^{[m]}$ into a left and right node we use the **risk reduction** achieved by that split:

$$\tilde{S}_{LR} = \frac{1}{2}\left[\frac{t_{\lambda_3}\left(G_{tL}^{[m]}\right)^2}{H_{tL}^{[m]}+\lambda_2} + \frac{t_{\lambda_3}\left(G_{tR}^{[m]}\right)^2}{H_{tR}^{[m]}+\lambda_2} - \frac{t_{\lambda_3}\left(G_t^{[m]}\right)^2}{H_t^{[m]}+\lambda_2}\right] - \lambda_1,$$

where the subscripts L and R denote the left and right leaves after the split.

LOSS MINIMIZATION - SPLIT FINDING II

Algorithm (Exact) Algorithm for split finding

Input I: *instance set of current node*
Input p: *dimension of feature space*
$gain \leftarrow 0$
$G \leftarrow \sum_{i \in I} g(\boldsymbol{x}^{(i)}), H \leftarrow \sum_{i \in I} h(\boldsymbol{x}^{(i)})$
for $j = 1 \rightarrow p$ **do**
    $G_L \leftarrow 0, H_L \leftarrow 0$
    for i in sorted(I, by x_j) **do**
        $G_L \leftarrow G_L + g(\boldsymbol{x}^{(i)}), H_L \leftarrow H_L + h(\boldsymbol{x}^{(i)})$
        $G_R \leftarrow G - G_L, H_R \leftarrow H - H_L$
        compute $\tilde{S}_{LR}$
    end for
end for
Output Split with maximal $\tilde{S}_{LR}$

